

Dear Editors,

I am pleased to submit the manuscript entitled "Robustness of Information Circulation under Entropy Constraints" for consideration for publication in *Physica Scripta*.

This manuscript develops an information-dynamical description of cyclic causal structures in finite complex systems. By formulating such structures as cycles of transfer entropy and reversed transfer entropy, the work shows that, under locally deterministic dynamics and non-synergistic interactions, a finite directed information-flow network requires circulation in order to sustain bounded nontrivial dynamics. The manuscript then analyzes how a pre-existing circulating information flow interacts with a newly activated external input at a shared receiving node with finite entropy capacity.

The main result is that, near the entropy-capacity limit of the receiving component, competing information flows become effectively exclusive: increasing the pre-existing circulating flow suppresses additional incoming transfer entropy. This provides an intrinsic information-theoretic mechanism for robustness under finite local entropy constraints. The analysis is supported by examples including Gaussian multiple-access channels, a multiplex-receiver model, and an AND-gate interaction, together with simulations of a binary stochastic network with sigmoid response.

I believe the manuscript is suitable for *Physica Scripta* because it addresses a general theoretical problem at the intersection of statistical physics, information theory, and complex-system dynamics. The results provide a compact framework for understanding how cyclic information flow can stabilize finite systems against external perturbations, with potential relevance to physical, biological, and engineered networks.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I have no conflicts of interest to declare. The data supporting the simulation results are openly available through the Zenodo repository cited in the manuscript.